

REWEAR-FUSED

A single molecular co-design loop that makes children's stretch clothing recyclable — by designing a new cleavable fiber and the matched enzyme that closes its loop. Neither existed before.

Engine A Cleavable Fiber

Engine B De Novo Enzyme

Engine C Recyclability Passport

Team of three · Stephen Sookra — Pravin — Vinh Le
72-hour build · In-person demo June 17, 2026 · Atlanta

DEEP-DIVE BRIEF
v2 · post-research

One loop, three engines, two prizes.

REWEAR-FUSED makes the unrecyclable recyclable. Children's stretch apparel can't go back into the clothing stream because of elastane — and we fix that not by attacking today's elastane, but by co-designing a new cleavable fiber together with the bespoke enzyme that breaks it down. A lock and key, designed for each other.

What we're proving

That AI can design *both* halves of a recycling solution at once — a fiber engineered to come apart, and a from-scratch enzyme engineered to take it apart — plus a working compliance layer that certifies a recovered garment is safe and recyclable.

Why it wins

It speaks to both decision-makers in the room: Carter's commercial pain (the track), and Cox's circular-economy investment thesis (the grand prize and the residency). The science is genuinely novel; the compliance layer is genuinely deployable.

ENGINE	ONE-LINE ROLE	OWNER
A · Fiber	Screen thousands of recipes to design a stretchy elastane with a breakable bond in an accessible spot.	PRAVIN
B · Enzyme	Design a protein from scratch that cuts exactly that bond, at a gentle pH that spares cotton.	PRAVIN
C · Passport	Classify and certify a garment against real US/EU chemical regulation; emit a scannable Digital Product Passport.	VINH LE
Experience	The 3D molecular views, the closed-loop animation, the dashboard, and the pitch narrative.	STEPHEN

THE NUMBERS THAT FRAME EVERYTHING

Elastane is in **~80% of clothing**, and just **1%** of it makes a recycler reject a whole garment. **92 million tonnes** of textiles are wasted yearly. The spandex market runs **~\$9B today toward ~\$20B by 2033**. And new law — California **SB 707** (\$50K/day penalties, 2026) and the EU **Digital Product Passport** (2027) — turns this from optional to mandatory.

A \$12 onesie can outlive everyone who ever wears it.

Children's stretchy clothes — onesies, leggings, sleepers — are almost impossible to recycle. The culprit is **elastane** (spandex), the fiber that makes clothing stretch. And the problem is more brittle than it sounds.

Elastane poisons the recycling stream

It appears in roughly **80% of all clothing**, and it doesn't take much: as little as **1% elastane** in an otherwise all-cotton garment is enough for a textile recycler to **reject the entire piece**. The stretchy fiber clogs and contaminates mechanical recycling machinery, soiling the output.

So today's "recycling" is downcycling

When Carter's collects worn baby clothes through **KIDCYCLE** (run with TerraCycle), it cannot make them into clothing again. Natural fibers become building insulation; synthetics and blends become composite lumber, mattress filler, and **pet bedding**. A onesie never becomes another onesie.

Why elastane specifically defeats recycling

Manufactured elastane is a **segmented polyurethane-urea**: hard crystalline blocks give it strength, soft amorphous coils give it stretch. Those crystalline domains are dense and chemically sealed — no natural enzyme can get inside, and mechanical shredding just smears the rubbery fiber through the cotton. Chemical recycling chokes on it too. Recent work confirms that even promising lab enzymes only touch commercial elastane after aggressive ball-milling and cellulase pre-treatment — evidence of just how strong the crystallinity wall is.

The scale, in numbers

92 Mt

of textile waste produced globally each year (UN Environment Programme) — roughly a garbage truck of clothing landfilled or burned every second.

~1%

of garments are recycled fiber-to-fiber today. Most is landfilled, incinerated, or downcycled into lower-value products.

18–26%

of gross textile waste *could* be captured by fiber-to-fiber recycling by 2030 if the technical barriers fall (McKinsey). Elastane is one of the largest barriers.

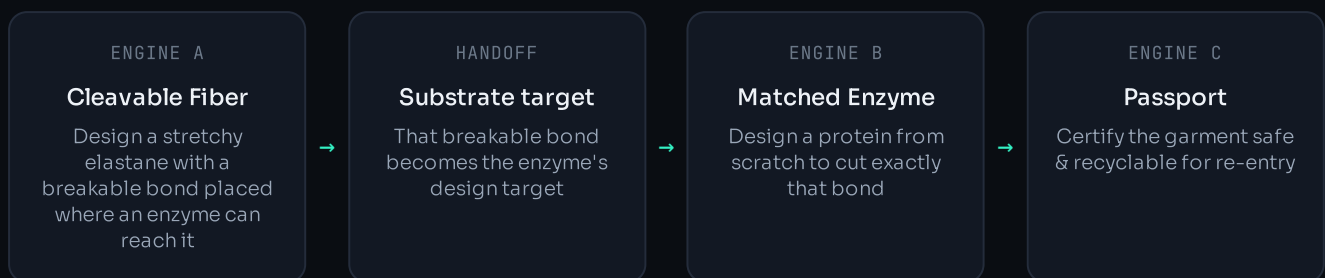
THE TWO WALLS BLOCKING CARTER'S

Wall 1 — chemical uncertainty: Carter's can't prove a recovered garment is chemically safe for infant skin, so it can't re-enter it into a supply chain. **Wall 2 — the elastane itself:** even if it were certified safe, the elastane can't be broken down. REWEAR-FUSED is built to knock down both.

Don't fight the fiber. Redesign both sides of the lock and key.

Every prior attempt tries to invent an enzyme that can chew through *today's* elastane. That fails, because manufactured elastane is physically sealed off. **REWEAR-FUSED inverts the problem.**

Instead of attacking an impossible target, we **co-design two matched molecules at once**: a brand-new elastane fiber engineered to be cleavable, and a brand-new enzyme designed from scratch to be the exact molecular scissors for that fiber. They are a **lock and key built for each other** — and neither exists in the world today.



This is what makes it **one system, not three projects**: the fiber design directly produces the enzyme's target, and the enzyme's requirements feed back as constraints on the fiber. The output is a **matched pair** plus the software that certifies the loop.

THE HONEST FRAMING — AND WHY IT'S A STRENGTH

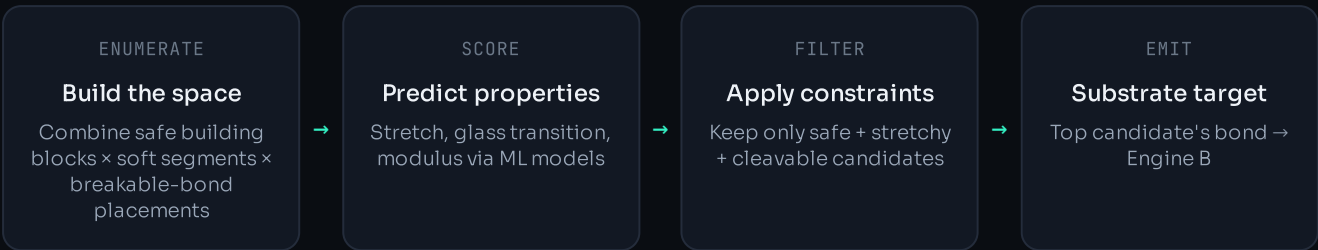
Everything we produce in 72 hours is an **in-silico design** — computer-designed molecules with benchmark-referenced confidence scores, not yet wet-lab-validated. We say so plainly, on stage. Designing the molecules is real and impressive; growing the enzyme and spinning the fiber is the forward vision — and it's exactly what the Cox Cleantech Residency exists to fund next. Over-claiming loses deep-tech rooms; calibrated honesty wins them.

A spandex engineered to come apart on command.

Engine A designs a new elastane that performs like the real thing — stretches past 400% and snaps back — but carries a deliberately breakable bond in a place an enzyme can actually reach.

The method: constraint-guided screening, not "AI invents a polymer"

This distinction matters, and a sharp judge will test it. Today's polymer-design AI models openly state in their own papers that they handle simple polymers — not the segmented block copolymers that elastane is. So we don't claim to generate a fiber from thin air. Instead we **enumerate** thousands of real, buildable recipes, **score** each with property-prediction models, and **filter** down to the few that are simultaneously stretchy, safe, and breakable.



The locked design rules (each is a defense against a likely judge question)

RULE	WHY IT'S NON-NEGOTIABLE
Safe (aliphatic) building blocks only	Aromatic alternatives break down into compounds linked to cancer — unacceptable next to a baby's skin. Aliphatic chemistry releases no such compounds. This is a hard safety gate.
Breakable bond in the soft, amorphous region	The crystalline hard blocks carry the load and the stretch-recovery. Put the cleavable bond in the accessible soft phase so the enzyme can reach it without destroying performance.
Hard-segment content 25–35%	Below ~17% it degrades too fast and loses strength; above ~40% it won't degrade at all. The 25–35% window is where stretch and recyclability coexist.
In-silico, benchmark-referenced	Cyclic fatigue and elastic recovery can't be predicted by today's ML — those need wet-lab validation. We show predicted properties with that caveat visible.

The tools

RDKit (enumerate candidate chemistries) · PolyID and a polyurethane-elastomer ML model (predict stretch and thermal properties, with published accuracy in the R^2 0.85–0.91 range on related elastomers). The frontend shows the screening funnel, candidate cards, and an honest stretch-vs-recyclability trade-off curve with the design window shaded.

MARKET DETAIL — THE FIBER SIDE

The global spandex market is roughly **\$9B in 2023, projected to ~\$20B by 2033** at about an 8% annual growth rate. The incumbents are already racing toward *bio-based* elastane (lower carbon), but **no major producer has shipped a recyclable elastane** — that's the open lane REWEAR-FUSED targets. Crucially, our fiber is designed to be a **drop-in** for existing dry-spinning manufacturing, so an incumbent could license it without re-tooling. Comparable new-fiber commercializations (bio-based stretch, certified-biodegradable stretch) have reached brands on ~1–3 year timelines, implying a realistic market horizon of **2030–2032**.

Open novelty flag (verified): no publicly disclosed AI-designed enzymatically-cleavable segmented polyurethane-urea elastane exists today.

A protein that never existed, designed to close the loop.

Engine B is the grand-prize centerpiece: a **carbamate hydrolase designed atom-by-atom from scratch** to grab and cut the exact bond in Engine A's fiber — the first of its kind.

The pipeline (the same class of toolchain behind 2025–2026 breakthroughs in *Science* and *Nature*)



How to read it in plain terms

Step 1 sculpts a protein shape with a pocket built to hold the fiber's breakable bond. Step 2 picks the chain of amino acids that will actually fold into that shape. Step 3 predicts the folded structure and checks the bond really sits in the pocket. Step 4 verifies the catalytic machinery — the chemical "blade" — is correctly preorganized. Step 5 compares the result against every known protein to confirm it's new. The output is a downloadable structure file the judges can inspect.

The locked design rules

RULE	WHY
Gentle, near-neutral-pH active site	The natural enzyme family this is inspired by only works in a harsh alkaline bath that would dissolve the cotton we're trying to recover. Designing from scratch lets us build a classic Ser-His-Asp active site that works at a gentle, cotton-safe pH.
Designed against Engine A's bond	The enzyme isn't generic — its pocket is shaped for the specific bond Engine A placed in the fiber. That's the "matched pair" that proves it's one integrated loop.
Novelty verified, not assumed	FoldSeek confirms there's no close match in the protein database (a low structural-similarity score = a genuinely new fold), which is what makes this a defensible, ownable result.

Compute & honesty

The whole design job runs on rented cloud GPUs for a few hundred dollars over the build weekend; results are saved as static files so the live demo never calls a GPU. We avoid non-commercial-licensed models so the result

stays commercially usable. **Real in 72 hours:** the designed structure, its confidence scores, and the novelty proof. **Forward vision:** expressing the enzyme in the lab and measuring its real-world cutting rate.

MARKET DETAIL – THE ENZYME SIDE

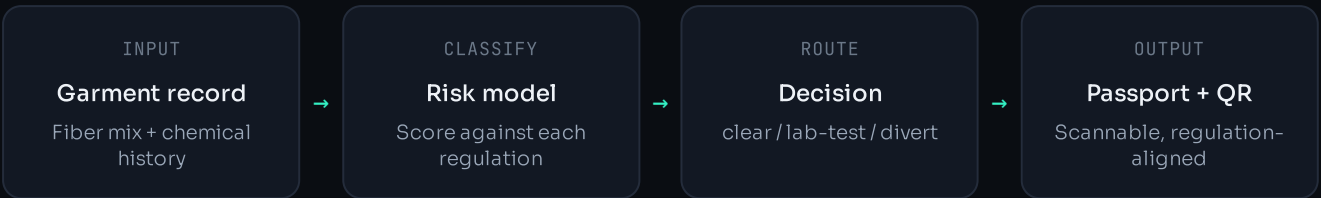
Enzymatic textile recycling is an emerging category with active commercial interest (e.g. enzyme-based polyester recycling has already attracted hundreds of millions in investment). A purpose-built enzyme that closes the loop on the single most problematic fiber — elastane — is a high-value, licensable asset for recyclers and fiber producers. Open novelty flag (verified): no publicly disclosed de-novo-designed carbamate hydrolase / urethanase exists today. That's the moat.

The deployable layer that makes the business case real.

Engine C is the piece a brand could use in Year 1, and the piece judges can **scan on stage**. It turns the project from "impressive science" into "Carter's saves money under the law."

What it does, in plain terms

A garment comes in. The system reads its fiber composition and known chemical history, checks it against real chemical-safety and recycling regulations, and decides one of three things: **clear** (safe to recycle), **lab-test** (needs a check), or **divert** (can't go back). Then it issues a clean **Digital Product Passport** — a scannable record of what the garment is made of, whether it's chemically safe, and how it should be recycled.



The three build blocks

BLOCK	WHAT IT IS
1 · Data backbone	Pull and clean public data: the U.S. consumer-product recall database, the EU rapid-alert safety system, and published PFAS-in-textiles datasets. These become the training examples and the "known-bad" reference set.
2 · The classifier	A gradient-boosted model (LightGBM) that scores a garment per regulation and outputs the clear / lab-test / divert decision with a probability. Because unsafe cases are rare, we report precision-recall performance — the honest metric — not plain accuracy, which would look misleadingly high.
3 · Passport + API	A FastAPI service that serves the classifier and generates the Digital Product Passport as structured data aligned to the EU passport format, with a scannable QR link. This is what the dashboard reads.

The regulations it encodes

CPSIA (lead, phthalates, children's-product flammability) · California AB 1817 & AB 347 (PFAS in textiles) · California SB 707 (textile producer responsibility) · New York and Maine PFAS laws · EU REACH restricted aromatic amines · OEKO-TEX Standard 100 Class I (the strict infant tier) · EU ESPR Digital Product Passport.

MARKET DETAIL — WHY THIS IS THE WEDGE

California SB 707 makes textile producers financially responsible for end-of-life, with penalties up to **\$50,000/day** and producers required to join the state's recycling organization by July 1, 2026. Fees are expected to be **eco-modulated** — lower for products designed to be genuinely recyclable. The **EU Digital Product Passport** for textiles (around 2027) will require exactly the disclosure this engine produces. So the Passport isn't a cost — it's a **fee-reducing compliance asset** that pays for itself.

It also reframes the microplastic question: enzymatic end-of-life replaces an ocean-shedding fate with a controlled industrial-hydrolysis fate.

Three people, three clean lanes, zero collisions.

The work splits so all three of us move in parallel without waiting on each other — and so the three engines add up to one coherent story.

STEPHEN SOOKRA

Frontend • Experience • Narrative

Builds the entire visual experience — the interactive 3D molecular views, the closed-loop animation, the compliance dashboard's interface, and the pitch narrative — tuned so heavy 3D still feels clean, not cluttered. The layer that makes the science visible and creates the "shouldn't be possible" reaction.

PRAVIN

Backend • The Molecular Pipelines (Engines A + B)

Owns the deep-tech core: the constraint-guided fiber screen and the from-scratch enzyme design pipeline. This is the grand-prize science — producing the matched fiber-and-enzyme pair that has never existed — delivered as pre-computed files the frontend renders.

VINH LE

Backend • Passport + Data/Infrastructure (Engine C)

Owns the deployable compliance layer and the data backbone: the regulatory datasets, the classifier, and the Digital Product Passport service. This is the demo floor — the guaranteed-working product that proves value even before the molecular science lands — and it's separable from the GPU-heavy work, so it can move fast and independently.

How the pieces connect

Each backend engine produces **pre-computed result files** (designs, scores, passports); the frontend reads those files and renders them. Critically, **the live demo never depends on a real-time computation** — everything is pre-baked, so the pitch physically cannot break on stage. That single rule is what makes a 72-hour deep-tech demo safe.

- **Pravin → Stephen:** fiber designs + enzyme structures (3D files + data) for the molecular views.
- **Vinh → Stephen:** garment classifications + passport data for the dashboard.
- **Pravin ↔ Vinh:** shared data conventions so the three engines tell one coherent story.

One pipeline, end to end.

From a worn garment to a certified, re-spinnable fiber — and from raw research to a demo that can't fail.



The data contract that keeps three people in sync

Everyone agrees on the shape of the files before building. The frontend is built against realistic mock files from hour one, then swaps in the real pipeline outputs as they land — same shapes, no rework.

PRODUCER	ARTIFACT	CONSUMER
Engine A (Pravin)	Fiber candidates + properties + the trade-off curve data + the substrate target for B	Frontend + Engine B
Engine B (Pravin)	Enzyme structure files + confidence scores + novelty verdict	Frontend
Engine C (Vinh)	Garment classifications + the Digital Product Passport JSON + QR link	Frontend
Frontend (Stephen)	Five views that read all of the above and tell the story	The judges

DEMO SAFETY — NON-NEGOTIABLE

Everything runs from pre-computed files, works offline on the demo laptop, and is backed by a recorded video of the full flow. If the venue Wi-Fi dies mid-pitch, we switch to the recording and never miss a beat. We bring two laptops and a phone hotspot. Nothing in the live demo depends on the network or a GPU.

The tailwind is already blowing.

Three forces make this the right idea at the right moment: a large and growing fiber market, a wall of new regulation, and an investor whose thesis we match exactly.

The market

\$9B →

spandex market today vs. projected ~2033 (~8% CAGR). Whoever cracks recyclable elastane owns a slice of this.

\$20B

92 Mt

annual textile waste; the value of raw materials lost to landfill/incineration is estimated near \$100B/year.

18–26%

of textile waste reachable by fiber-to-fiber recycling by 2030 if barriers like elastane fall (McKinsey).

The regulation

RULE	WHAT IT FORCES
California SB 707	First U.S. textile producer-responsibility law. Producers join the state recycling org by July 1, 2026; penalties up to \$50,000/day; fees expected to drop for recyclable-by-design products.
EU Digital Product Passport	Textiles passport arriving ~2027: brands must disclose composition, chemical safety, and end-of-life — exactly what Engine C produces.
EU separate textile collection	Mandatory across the EU since Jan 1, 2025 — more recovered textile volume that needs recyclability certification.
PFAS-in-textiles laws	California, New York, and Maine restrict "forever chemicals" in apparel — raising the bar for chemical certification that Engine C automates.

The investor fit

"We're the Nexus Circular of textiles."

Cox Cleantech led a **\$150M round to become majority owner of Nexus Circular** — proprietary chemistry that turns hard-to-recycle plastic into virgin-equivalent feedstock for global brands. REWEAR-FUSED is the textile analog of a bet they've already placed at nine figures. Cox has committed to circular-economy and zero-waste goals and runs a residency that explicitly wants

teams with **progress toward intellectual property** and a **near-complete product ready to pilot**.
We arrive looking like that company.

Five minutes. One undeniable demo. One line they remember.

TIME	BEAT
0:00–0:20	Hook. Hold up a Carter's onesie: "This \$12 onesie will outlive everyone in this room — in a landfill, because of 1% of one fiber."
0:20–0:50	Problem + stakes. Elastane in 80% of clothing; 1% rejects a whole garment; SB 707 fees in 2026; Carter's HQ is minutes away.
0:50–2:30	Solution + live demo. The fiber design, the matched enzyme structure, and a live scan of the Digital Recyclability Passport on stage.
2:30–3:30	Business case. \$9B→\$20B market; Carter's avoids SB 707 fees and pre-positions for the EU passport; the pilot ask.
3:30–4:30	Why us + the ask. "We're the Nexus Circular of textiles" — and we're applying to the Cox Cleantech Residency.
4:30–5:00	Close. One memorable line. Stop. Don't over-explain.

"Carter's sells the first clothes a person ever wears. We make sure they're not also the first clothes that outlive them."

Pre-empting the hard questions (each answer is built into the product)

- **"What toxic byproducts come off?"** — none; the fiber uses safe aliphatic chemistry, shown as a green check on the passport.
- **"Won't the enzyme's harsh pH ruin the cotton?"** — no; we designed a gentle near-neutral active site, labeled on the enzyme view.
- **"Won't making it cleavable kill the stretch?"** — no; the breakable bond sits in the soft phase, and the trade-off curve shows the design window.
- **"Did you actually make any of this?"** — these are in-silico designs with confidence scores; wet-lab validation is the funded next step. Honesty as armor.
- **"Will Carter's pay for this?"** — they already pay to downcycle into pet bedding; we replace that line item with a fiber-to-fiber loop plus a fee-reducing compliance asset.

What we walk out with, if we win

- **The track** — the only entry that closes both of Carter's recycling walls and hands them a compliance asset.
- **The grand prize** — a genuinely new enzyme, designed in 72 hours, that shouldn't be possible.

- **The residency interview** — we arrive looking like a fundable company mapped onto a bet Cox already made.

ONE HONEST NOTE FOR THE TEAM

The exact judging rubric isn't published — it's behind the event registration. Everything here is built to match the most likely criteria (impact, innovation, business viability, presentation), but the one thing only we can do is register and ask the organizers directly for the rubric. Let's do that early.

*Neither the molecule nor the enzyme existed 72 hours ago.
Let's go build the thing that does.*